

DPL: a tool for assessing a district on sustainability

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1. INTRODUCTION

Local governments in the Netherlands are developing policies to plan districts in a sustainable way. The elaboration of these policies is not a simple task. One of the reasons is the broad and confusing definition of sustainability. Within this broad definition there is a considerable chance for incompatible goals. Various parties use different interpretations for the concept of sustainability. For parties involved, such as local authorities and property developers, it is not easy to jointly formulate unambiguous goals for sustainable planning. The definition of sustainability for an urban plan can not afford to be unclear because it has to fight its way among a large set of other requirements.

In the last ten years initiatives started in the Netherlands to develop instruments for assessing sustainability on the level of districts. Examples of Dutch tools are: de Bestuurlijke Bandbreedte Methode (Utrecht), de Stads Stolpmethode (Amsterdam) and the Ecopolis Strategie (Nijmegen). However, the interpretation of sustainability in these instruments varies strongly. The Councils of Housing, Spatial Planning and the Environment concluded that due to the diversity and nature of the instruments it is insufficiently clear what is sustainability, who makes the choices about sustainability of a plan and based on what.

Against this background, the initiative ‘Sustainability-Profile for Districts’ (the Dutch acronym for Duurzaamheid Prestatie voor een Locatie is DPL) started in May 2000. The DPL initiative was initiated by a consortium of the Dutch Ministry of Housing, Spatial Planning and the Environment – Directorate General of Housing, IVAM Environmental Research, TNO – Building and Construction Research and TNO Environment, Energy and Process Innovation (J.G.M. Kortman et al., 2001).

2. AIM OF THE INITIATIVE

The initiative aims to develop and test a challenging tool for assessing sustainability at district-level. The target groups for the application of the tool are municipalities, project developers, citizens, business community’s etc. The name of the tool is DPL. The tool assesses in a clear and transparent way the spatial plan for a district on sustainability, based on the information from the urban plan. It so helps urban designers to creatively improve the sustainable performance of a district.

The tool can be used in selected phases of an urban planning process. In the phase of definition the tool supports to formulate quantifiable objectives for sustainability as part of a total set of requirements. In the phase of design it helps to compare the different plan

alternatives and improves the urban plan. In the phase of evaluation the tool can be used to monitor a location and compare the sustainability performance of various districts in a town.

Until now, the initiative performed three steps:

1. Define and unravel the concept of sustainability for urban planning;
2. Develop a prototype instrument to assess a district on sustainability;
3. Test this prototype in three case studies.

The outcomes of these three steps are presented in the next paragraphs.

3. DEFINE AND UNRAVEL SUSTAINABILITY FOR URBAN PLANNING

In the Netherlands some first-generation instruments, such as the Bestuurlijke Bandbreedte Methode are already tested in practice. These outcomes of these Dutch instruments still vary considerably. The ambition of the DPL project is to make a next step and develop a second-generation instrument that sets a standard for the quantitative determination of the sustainability performance of a district with a consistent set of criteria. For some criteria it appeared not yet possible to quantify the impacts. The tool therefore will allow to add qualitative results for those criteria where quantification is not possible.

The instrument is built upon three axes (see Figure 1). The X-axis represents the concept of sustainability, the Y-axis represents the physical feature of a district and the Z-axis the planning process. In various phases of the planning process the instrument helps to determine the sustainability performance of a district from the physical features of the district.

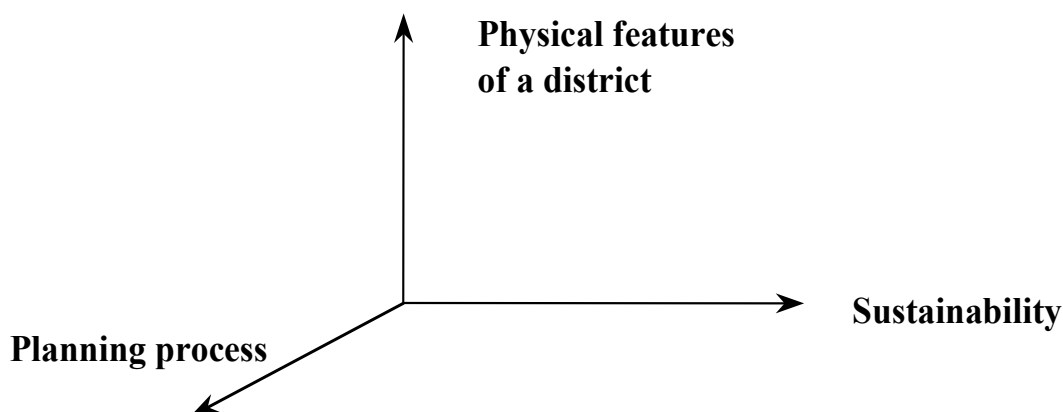


Figure 1 The 3 axes of the DPL instrument

Sustainability on the X-axis is interpreted as an integral concept, which is systematically divided into various levels of criteria, which are presented in the form of a pyramid. The top is presented by the overall concept of sustainability (see Figure 2). On the second level you find three elements which are closely connected to the three P's: environment (Planet), liveability (People) and economy (Profit).

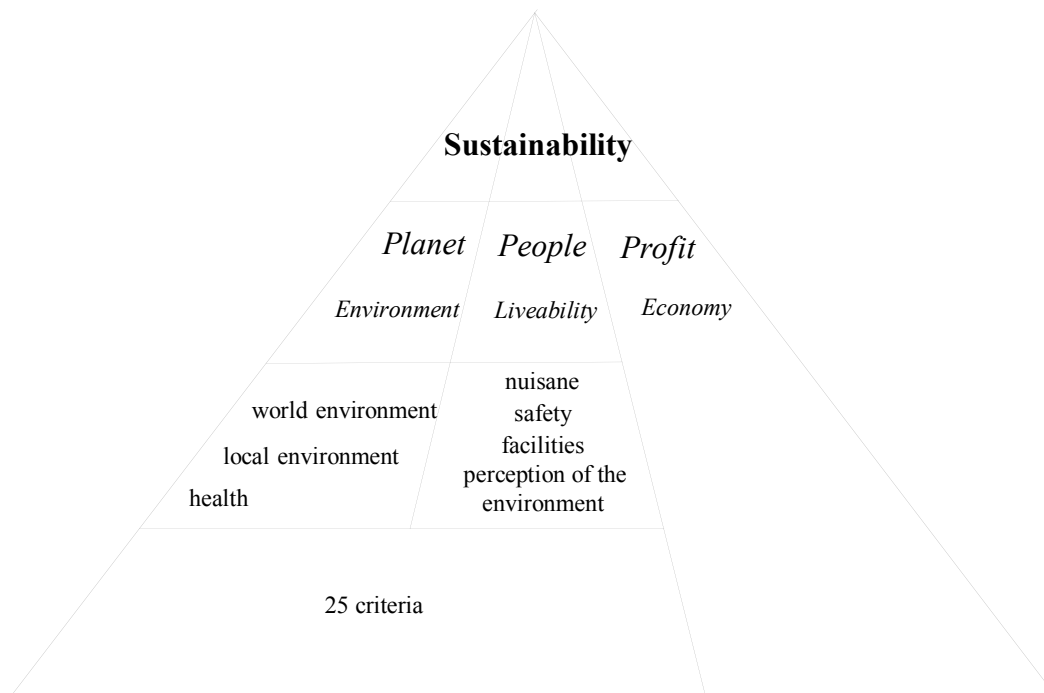


Figure 2 Pyramid for sustainability

On the second level the two elements, Environment and Liveability are unravelled in 7 themes (see Figure 2), followed by 25 criteria in the fourth level. These 25 criteria are selected for the prototype tool from a total of 200 criteria that were found in literature and currently existing tools. This selection is meant to be a first set towards a more complete and consistent set of criteria for sustainability. Each criterion is connected to an indicator. The DPL scores for each criterion are determined by translating information from the urban plan with a formula and data into a score for an indicator. The scores for these indicators are the outcome of the DPL tool.

4. DEVELOPING THE DPL TOOL

On the basis of this pyramid for sustainability a first DPL prototype tool was developed that assists planners to determine the sustainability performance of a district. It consists of three parts:

1. The DPL questionnaire;
2. The DPL matrix;
3. The DPL profile.

4.1 The DPL Questionnaire

The sustainability performance of a district is determined in the prototype on the basis of the physical characteristics of a neighbourhood. The necessary information of an existing neighbourhood or information from an urban plan is filled in the questionnaire. Examples of physical characteristics are the number and floor area of houses, offices, roads, the modal split and nuisance from noise, odour etc. This information is automatically transformed to the DPL matrix.

4.2 The DPL Matrix

The DPL matrix automatically transforms the information from the questionnaire into indicator scores by using the formula and the data. The DPL consists of two axes (see Figure 1). The Y-axis presents the various functions of a neighbourhood, such as houses, parks, offices, infrastructure, factories etc. The X-axis presents the 25 indicators for sustainability. Each box of the matrix consists of data and formula. For example for the indicator energy use of houses, data for energy consumption is connected to the m² living space, which is filled in the questionnaire.

4.3 The DPL profile

With help of the matrix the DPL profile is calculated and related to the amount of ground surface of the district in question (see Figure 4). In this figure the DPL profile is limited to 8 indicators.

5. TESTING THE DPL TOOL IN THREE CASE STUDIES

In the first DPL project the practical value of the tool was tested in order to know if the instrument is suitable for the needs of the Dutch planning-practice. Therefore the tool was tested in three areas that are being reconstructed:

- Geuzenveld, Amsterdam (district with low and middle high rising buildings, which was firstly constructed after the second World War);
- Poptahof, Delft (district with low, middle and high rising buildings, which was firstly constructed in the sixties 2001);
- Transvaal, The Hague (district with middle high rising building, firstly constructed at the end of the 19th century).

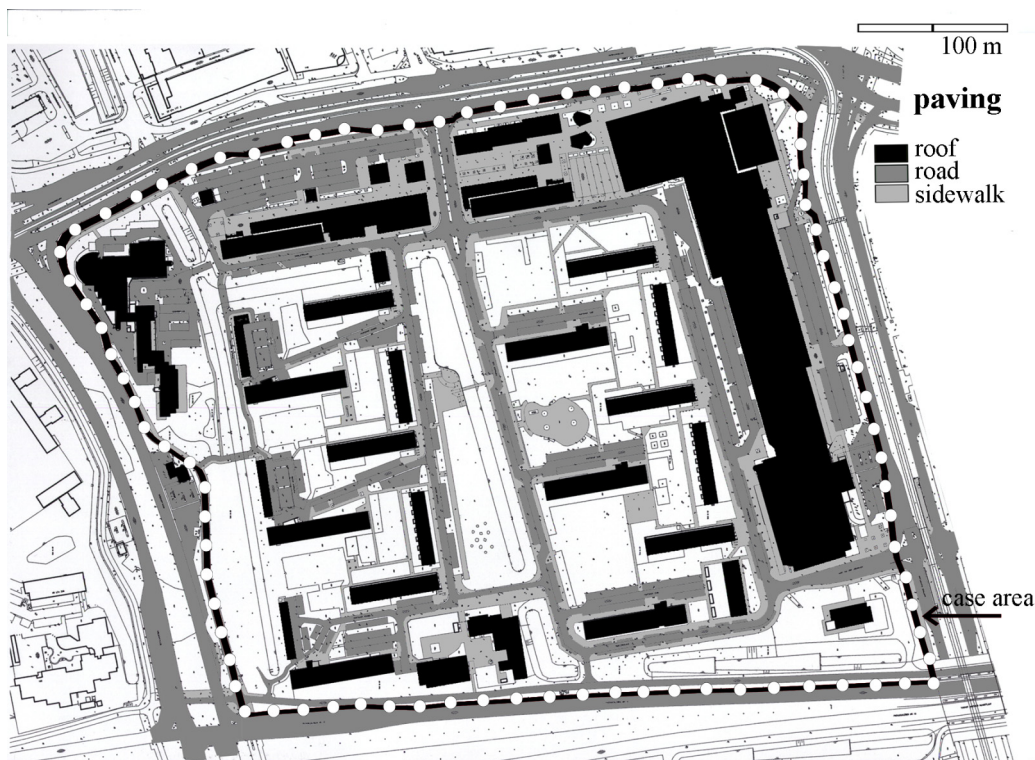


Figure 3 One of the case studies: Poptahof, Delft

The case studies were focused on the practical side and intrinsic value of the instrument: Is it manageable, conveniently arranged, quickly to apply; does it meet the needs of the parties; is there sufficient data available; is it sufficiently capable of mapping the actual practical value, perception and appreciation? The members from municipalities (Amsterdam and The Hague) and a research group from Delft collected the data from these neighbourhoods and transported these data into the questionnaires. Based on the (sometimes incomplete) set of data the sustainable profiles of the three neighbourhoods were calculated by IVAM with help of the DPL matrix. A part of the sustainable profile is presented in Figure 4. Because it was the first tryout there were not yet reference scores available for various types of districts in the DPL tool. The indicator scores for each of the three districts are compared to the average score (=100%) of the three neighbourhoods.

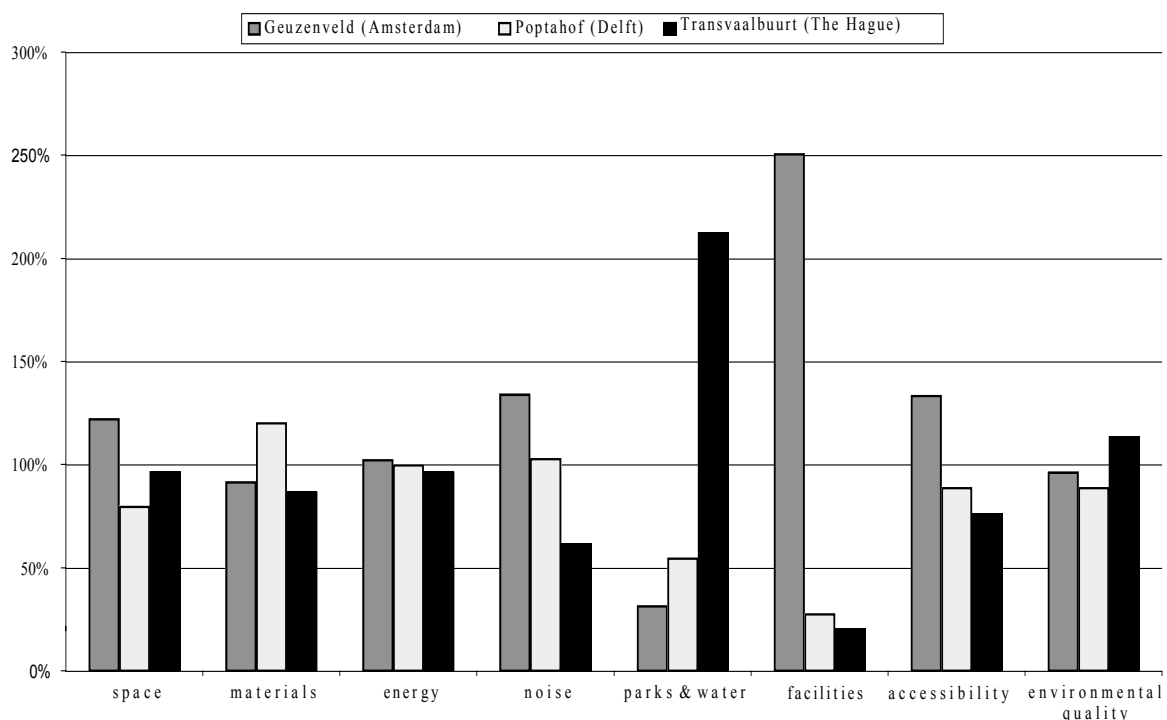


Figure 4 Comparing the DPL profiles of 3 districts

The lower the scores the better the performance of the district. The profile indicates that the district Poptahof in Delft has relatively good scores on the aspects of 'space' and 'facilities'. The oldest district Transvaal in The Hague has relatively good scores on 'space', 'materials', 'noise', 'facilities' and 'accessibility'. The district Buurt 9 in Amsterdam scores well on the criteria 'parks' and 'water'. Comparing the three districts with various typologies has a limited value. Therefore the scores from the first three cases has to be treated very carefully because there are not yet good reference profiles for various types of districts.

6. PRELIMINARY CONCLUSIONS

Municipals were able to deliver about 75% of the necessary information. With help of the DPL prototype tool it was possible to determine (part of) the DPL profiles of the three districts. The comparison of the three cases can not be used to select the most sustainable district. The scores from the first three cases has to be treated very carefully, because there are not yet good reference profiles for various typologies of neighbourhoods. The comparison is only performed to test the usefulness of the instrument. Especially for the element of liveability necessary information is lacking. Additional information should be used in the case

of an existing neighbourhood from liveability research by questionnaires filled in by the inhabitants of a district.

Sustainable building on district level is still in its infancy. Local governments still lack a comprehensive view of how to improve districts in a sustainable way. The instrument DPL is a guide to find the most efficient improvements.

7. REFERENCES

Kortman, J., H. van Ewijk, P. van Konijnenburg, R. Lanting, A. de Groot-van Dam, R. Kleefman and F. Timmermans. 2001. Duurzaamheidsprofiel van een locatie, Ontwikkeling en test van het DPL instrument versie 1.0. IVAM Environmental Research, TNO - Building and Construction Research and TNO - Environment, Energy and Process Innovation. Amsterdam, Delft.